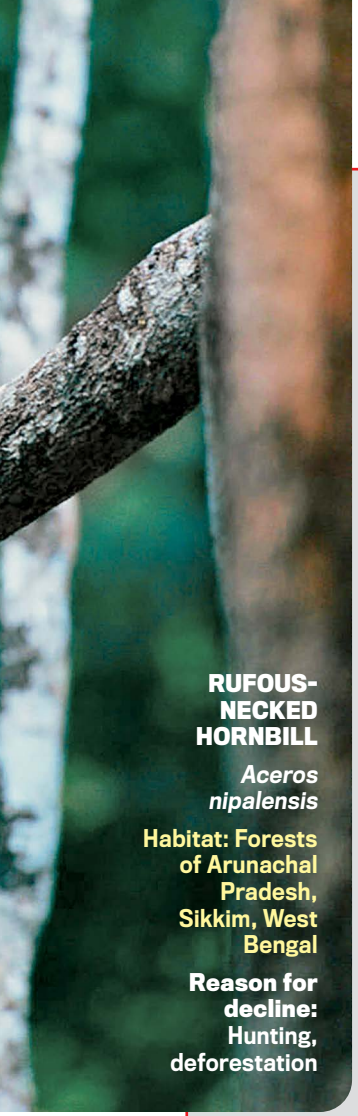


RUFOUS-NECKED HORNBILL

Aceros nipalensis

Habitat: Forests of Arunachal Pradesh, Sikkim, West Bengal

Reason for decline: Hunting, deforestation

SILENCE OF THE HORNBILL

TREND: India is home to 9 species of hornbills. Of these, the Malabar grey hornbill, found in the Valparai plateau of the Western Ghats, declined by 56 per cent between 2004 and 2018. The Rufous-necked hornbill and Wreathed hornbill are declining.

WHY: Hornbills use the same nest year after year. Their peculiar nesting behaviour makes them particularly susceptible to disruption. Hunting poses another threat to the Great hornbill, the Rufous-necked hornbill and the Wreathed hornbill in parts of the eastern Himalaya. Forests are being lost to illegal logging, land use is changing and monoculture plantations are thriving. In northeast India, only 5 per cent landscape is suitable for these birds.

only in India's Eastern Ghats, is listed as critically endangered in the IUCN red list and was last seen in 2004. But even though it hasn't been sighted in many years, a small population is believed to be thriving somewhere in the less-explored scrub forests of Andhra Pradesh.

WHY DO WE NEED BIRDS?

Birder Das finds being with birds meditative. "Because we are so far removed from wild birds today, we have forgotten how important they can be for our peace of mind," he says. "I credit the sparrows and parrots in our local park to my success at work," echoes birdwatcher and banking executive Abhishek Chawla, 47, from Mumbai. "Every morning, I go to hear them and calm my nerves. That morning wake-up routine is extremely important for my well-being." A 2022 research in *Scientific Reports* supported the idea that birdsong reduces anxiety and paranoia not only among those who reported depression but is beneficial even for 'healthy' humans.

Not just human health, birds are crucial for the well-being of the entire ecosystem. "If birds can't survive," says Kartick Satyanarayan, CEO and co-founder of Wildlife SOS, "it won't be long before humans will find it hard to survive too. Birds like sparrows are indicator species that reflect the health of the ecosystem." Adds Menon, "Birds are important for their role in dispersal of seeds, pollination and are key predators of a number of invertebrates that could reach pest proportions without birds. They are also a great indicator of climate change, as their migratory patterns are dependent on climate patterns."

So, while you may find the vulture an ugly bird, the scavenger helps prevent outbreak of disease by consuming the decaying carcasses in the plains where it's typically found in India. The decline in the vulture population has long been linked with the alarming increase in the population of stray dogs, especially in urban areas, across India, and the rise in cases of rabies.

BIRDS ARE ALSO ESSENTIAL for the survival of local ecosystems. In the Western Ghats, for example, most endemics, such as the White-bellied treepie and the Wayanad laughingthrush, are inhabitants of the rainforest, and are key to its survival, which, in turn, influences the weather, soil and landscape of the region. South Asia is home to 232 species of endemic birds, of which 30 are endemic to the Andaman & Nicobar Islands, 36 to the Western Ghats and Sri Lanka, and 95 to the Indian subcontinent as a whole. The SoIB report shows a more than 50 per cent decline in the abundance of endemic birds in the Western Ghats and Sri Lanka between 2014 and 2022. "We don't realise it, but birds serve an important ecological function and are important for our future," says Rushikesh Chavan, head of The Habitats Trust, a wildlife conservation body.

Other birds in rapid decline include open-habitat species like bustards and coursers, riverine sandbar-nesting birds like skimmers and some terns, coastal shorebirds, open-country raptors and ducks. In fact, the discovery that common species like ducks are also in decline came as a surprise and an alarm to experts.

But, what is killing our birds? Climate change, for one. In the summer of 2022, Gujarat witnessed the curious spectacle of birds falling from the sky because the extreme heat left them dehydrated and exhausted. Similar cases were reported from other parts of North India as well. According to an Intergovernmental Panel on Climate Change (IPCC) estimate, global mean temperatures have increased by 0.8°C in the past 150 years. Warming climes are affecting the migratory patterns of birds, causing them to move northwards or to higher elevations, in search of cooler environs. According to a March 2023 study by four Indian researchers in the journal *Diversity*, as many as 66-73 per cent of the 1,091 species they studied would shift northward or upward by 2070.

Climate change also possibly affects avian nesting, breeding and foraging behaviour. In a 2018 study